

An energy company is considering an investment in wind-power production. To determine whether this investment is economically feasible, the company seeks to estimate the expected revenue of a wind-power producer operating in a competitive spot market.

Consider a day-ahead spot market with five electricity producers. Three producers generate electricity from conventional sources, such as coal, whereas the remaining two rely on solar and wind power, respectively. The market is cleared once each day for 24 hourly trading intervals of the following day. For each interval, every producer submits a bid that specifies the electricity quantity it is willing to supply (MWh), together with the minimum price it is willing to accept ($\$/\text{MWh}$). The market operator then applies a merit-order clearing mechanism to determine both the market-clearing price and the dispatch schedule, subject to the requirement that total electricity demand be satisfied. If a producer's bid is accepted, the producer is obligated to supply the nominated quantity and is remunerated at the market-clearing price. If the bid is not accepted, the producer receives no revenue for that interval.

The energy company intends to develop a model to estimate the expected revenue of the wind-power producer on any future day by determining the hourly bid quantities and prices under uncertainty. On this basis, the model can subsequently be extended to examine outcomes across a range of short-term and long-term scenarios, thereby supporting an assessment of the investment's economic viability.

Please formalise this problem as an agent-based model, and specify the key components, including time scales, agents, state variables, control variables, processes, objectives, constants, uncertain parameters and stochastic variables. At this stage, the model should be limited to simulating single-day market operations.

Show less

Agent-Based Model: Day-Ahead Wind-Power Revenue Estimation

1. Overview

The model simulates a single day of operations in a day-ahead electricity spot market. Five producer agents submit bids for 24 hourly intervals. A market operator agent clears each interval via a merit-order mechanism. The primary output is the wind producer's total daily revenue, which serves as a sample from the distribution of possible revenues under uncertainty.

2. Time Scales

Scale	Description
Simulation horizon	One day (the "target day"), indexed d
Trading intervals	24 hourly slots, indexed $t \in \mathcal{T} = \{1, 2, \dots, 24\}$
Bid submission	A single event occurring before the target day begins; all bids for all 24 intervals are submitted simultaneously
Market clearing	One clearing event per interval t , executed sequentially after bid submission

3. Agents

3.1 Producer Agents

There are five producer agents, partitioned by technology type.

Conventional Producers $i \in \mathcal{C} = \{1, 2, 3\}$

These agents operate dispatchable coal-fired plant. Their offered quantity is bounded by physical capacity and their bid price reflects short-run marginal cost (fuel, operations and maintenance), which is known with reasonable certainty.

Solar Producer $i = S$

This agent's offered quantity is constrained by solar irradiance, which is weather-dependent and uncertain at the time of bidding. Its marginal cost of generation is near zero, so its bid price is typically set at or near zero to ensure dispatch.

Wind Producer $i = W$

This is the focal agent whose revenue the model is designed to estimate. Like the solar producer, its offered quantity is constrained by an uncertain, weather-driven resource. Its marginal cost is near zero. Its bid quantity and price constitute the primary decision variables of interest.

3.2 Market Operator Agent

A single, rule-following agent that collects all bids, applies the merit-order clearing mechanism for each interval, and returns dispatch decisions and clearing prices. It has no strategic behaviour and no objectives beyond satisfying demand at minimum cost.

4. State Variables

State variables describe the condition of the system at any point during the simulation.

Symbol	Domain	Description
$q_{i,t}^{\text{bid}}$	$\mathbb{R}_{\geq 0}$	Quantity bid by producer i for interval t (MWh)
$p_{i,t}^{\text{bid}}$	$\mathbb{R}_{\geq 0}$	Minimum price bid by producer i for interval t (\\$/MWh)
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price for interval t (\\$/MWh)
$x_{i,t} \in \{0, 1\}$	Binary	Dispatch indicator: $x_{i,t} = 1$ if producer i 's bid is accepted at interval t
$q_{i,t}^{\text{disp}}$	$\mathbb{R}_{\geq 0}$	Quantity dispatched from producer i at interval t (MWh); equals $x_{i,t} \cdot q_{i,t}^{\text{bid}}$
R_W	$\mathbb{R}_{\geq 0}$	Total daily revenue of the wind producer (\$)

5. Control Variables

These are the decision variables that the wind producer (and by symmetry each producer) chooses at bid submission time.

Symbol	Description
$q_{W,t}^{\text{bid}}$	Quantity the wind producer offers to supply in interval t (MWh); bounded above by the uncertain available wind capacity $\tilde{A}_{W,t}$
$p_{W,t}^{\text{bid}}$	Minimum price the wind producer is willing to accept in interval t (\\$/MWh)

At this stage of model development, the bids of all other producers are treated as exogenous inputs drawn from their respective distributions. In subsequent extensions, the control variables of competing agents may be endogenised to capture strategic interaction.

6. Processes

6.1 Resource Realisation

Before bids are submitted, each stochastic availability variable is drawn from its distribution:

$$\tilde{A}_{W,t} \sim F_{W,t}, \quad \tilde{A}_{S,t} \sim F_{S,t}, \quad \forall t \in \mathcal{T}$$

These realisations represent the maximum physically available generation capacity for the target day.

6.2 Bid Formation

Each producer submits a bid $(q_{i,t}^{\text{bid}}, p_{i,t}^{\text{bid}})$ for every interval t . The feasibility constraint for quantity is:

$$0 \leq q_{i,t}^{\text{bid}} \leq \bar{Q}_i \quad \forall i \in \mathcal{C}$$

$$0 \leq q_{W,t}^{\text{bid}} \leq \tilde{A}_{W,t}, \quad 0 \leq q_{S,t}^{\text{bid}} \leq \tilde{A}_{S,t}$$

6.3 Merit-Order Market Clearing

For each interval t , the market operator executes the following procedure:

1. **Sort** all bids in ascending order of bid price to form the aggregate supply stack.
2. **Traverse** the stack, accepting bids in order of price until cumulative dispatched quantity meets or exceeds demand D_t .
3. **Set** the clearing price λ_t equal to the bid price of the marginal (last accepted) producer.
4. **Dispatch** all accepted producers at their full bid quantity; reject all others.

Formally, let σ_t be the permutation that sorts producers by $p_{i,t}^{\text{bid}}$ in non-decreasing order. Define the marginal producer index m_t as:

$$m_t = \min \left\{ k \mid \sum_{j=1}^k q_{\sigma_t(j),t}^{\text{bid}} \geq D_t \right\}$$

Then:

$$\lambda_t = p_{\sigma_t(m_t),t}^{\text{bid}}$$

$$x_{i,t} = \begin{cases} 1 & \text{if } p_{i,t}^{\text{bid}} \leq \lambda_t \\ 0 & \text{otherwise} \end{cases}$$

Ties at the margin are broken by a pre-specified rule (e.g., pro-rata allocation or random selection).

6.4 Revenue Calculation

The wind producer's revenue for interval t is:

$$r_{W,t} = x_{W,t} \cdot q_{W,t}^{\text{bid}} \cdot \lambda_t$$

Total daily revenue:

$$R_W = \sum_{t=1}^{24} r_{W,t}$$

7. Objectives

Agent	Objective
Wind producer	Maximise expected daily revenue $\mathbb{E}[R_W]$, subject to the constraint that $q_{W,t}^{\text{bid}} \leq \tilde{A}_{W,t}$
Solar producer	Analogous to wind; maximise expected daily revenue
Conventional producers	Maximise profit; bid price set to recover marginal cost
Market operator	No optimisation objective; executes the merit-order rule deterministically

Agent	Objective
Modeller	Estimate the distribution of R_W to assess investment viability

8. Constants

These quantities are fixed for all simulation runs of a given scenario.

Symbol	Description
$\bar{Q}_i$	Installed capacity of conventional producer $i \in \mathcal{C}$ (MW)
$\bar{Q}_W$	Installed nameplate capacity of the wind producer (MW)
$\bar{Q}_S$	Installed nameplate capacity of the solar producer (MW)
c_i	Short-run marginal cost of conventional producer i ($\backslash$ \$/MWh); determines bid price floor
$n_C = 3$	Number of conventional producers
$\$$	$\backslash$ mathcal{T}

9. Uncertain Parameters and Stochastic Variables

A distinction is drawn between **uncertain parameters** (quantities that are fixed for a given day but unknown at bid submission time) and **stochastic variables** (quantities whose randomness is intrinsic to the process being modelled).

9.1 Uncertain Parameters

Symbol	Description
D_t	Electricity demand in interval t (MWh); determined by consumer behaviour and weather, unknown at bid time
c_i^{eff}	Effective marginal cost of conventional producer i , which may vary with fuel spot prices

9.2 Stochastic Variables

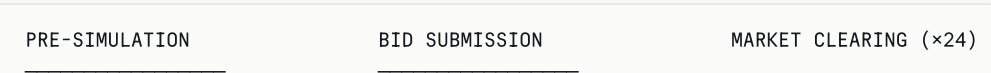
Symbol	Description
$\tilde{A}_{W,t}$	Available wind capacity in interval t (MWh); driven by wind speed, modelled as a stochastic process with temporal autocorrelation across t
$\tilde{A}_{S,t}$	Available solar capacity in interval t (MWh); driven by irradiance, exhibiting a strong diurnal pattern with stochastic perturbations
$\tilde{q}_{i,t}^{\text{bid}}, \tilde{p}_{i,t}^{\text{bid}}$	Bids of competing agents, treated as random variables when competitors' strategies are not explicitly modelled

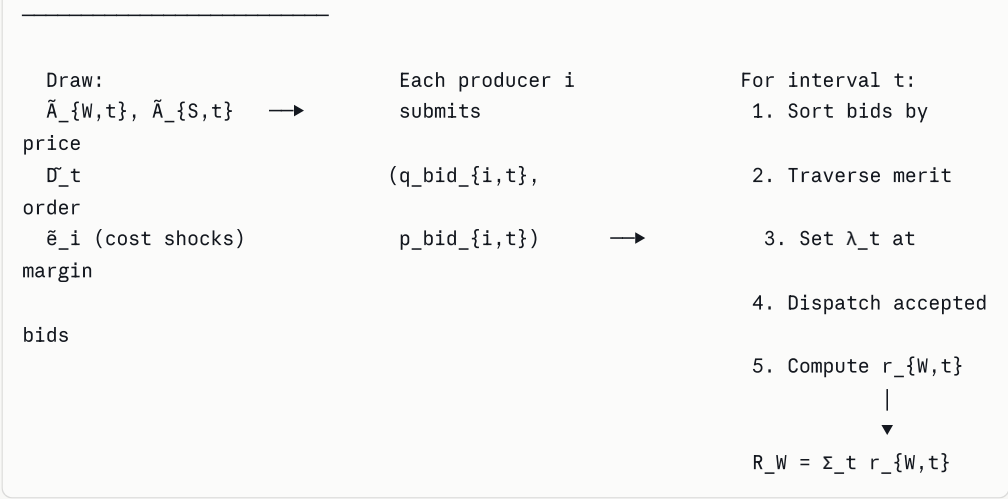
9.3 Dependence Structure

The following dependencies are material to model validity and must be specified:

- $\tilde{A}_{W,t}$ exhibits **serial autocorrelation** across intervals (wind persists over hours) and may be modelled as an autoregressive process or sampled from historical wind profiles.
- $\tilde{A}_{S,t}$ is **conditionally deterministic** given cloud cover: it follows a known diurnal curve scaled by a stochastic weather factor.
- D_t and $\tilde{A}_{W,t}$ may be **negatively correlated** in some markets (e.g., high wind coincides with low demand at night), which is consequential for revenue estimation.
- Competitor bids may be **correlated** with demand and availability through common weather drivers.

10. Summary Schematic





11. Model Extensions (Deferred)

The following components are noted for future development but excluded from the single-day model:

- **Multi-day simulation** with learning and adaptive bidding strategies
- **Seasonal and inter-annual resource variability** for long-run revenue distributions
- **Strategic bidding** by competitors (game-theoretic formulation)
- **Transmission constraints** and network topology
- **Capacity payments** and ancillary service markets
- **Carbon pricing** effects on conventional marginal costs

You can assume that the market demand follows a normal distribution with constant mean and variance. Please refine your specification accordingly and present the revised section.

9. Uncertain Parameters and Stochastic Variables (*Revised*)

A distinction is drawn between **uncertain parameters** (quantities that are fixed for a given day but unknown at bid submission time) and **stochastic variables** (quantities whose randomness is intrinsic to the process being modelled).

9.1 Uncertain Parameters

Symbol	Description
c_i^{eff}	Effective marginal cost of conventional producer i , which may vary with fuel spot prices

9.2 Stochastic Variables

Symbol	Description
D_t	Electricity demand in interval t (MWh)
$\tilde{A}_{W,t}$	Available wind capacity in interval t (MWh)
$\tilde{A}_{S,t}$	Available solar capacity in interval t (MWh)
$\tilde{q}_{i,t}^{\text{bid}}, \tilde{p}_{i,t}^{\text{bid}}$	Bids of competing agents, treated as random variables when competitors' strategies are not explicitly modelled

9.3 Demand Distribution

Hourly demand is modelled as an independent normal random variable with constant mean and variance, identical across all intervals:

$$D_t \sim \mathcal{N}(\mu_D, \sigma_D^2), \quad \forall t \in \mathcal{T}$$

where $\mu_D > 0$ is the mean demand (MWh) and $\sigma_D^2 > 0$ is the variance, both treated as constants of the model. The parameters μ_D and σ_D are added to the constants table in Section 8.

Three modelling assumptions follow from this specification and should be noted explicitly:

Independence across intervals. Demand realisations D_t and $D_{t'}$ are assumed independent for $t \neq t'$. This abstracts away the serial autocorrelation that is typically present in real load profiles (e.g., demand in hour 14 is correlated with demand in hour 13). The assumption is adopted for tractability at this stage and may be relaxed in subsequent extensions.

Support truncation. The normal distribution places positive probability on $D_t \leq 0$, which is physically inadmissible. In practice this probability is negligible provided $\mu_D \gg \sigma_D$. Where this condition holds, the distribution is used without truncation. If it does not hold, a truncated normal distribution $\mathcal{N}^+(\mu_D, \sigma_D^2)$ restricted to $\mathbb{R}_{>0}$ should be substituted.

Homogeneity across intervals. Constant mean and variance implies no diurnal demand pattern — the model does not distinguish peak from off-peak hours. This is a deliberate simplification for the single-day model. Extensions incorporating a time-varying mean $\mu_{D,t}$ (e.g., specified as a deterministic 24-hour load curve) and heterogeneous variance $\sigma_{D,t}^2$ would restore realism at the cost of additional parameters.

9.4 Dependence Structure

With demand now fully specified, the following stochastic dependencies remain material:

- $\tilde{A}_{W,t}$ exhibits **serial autocorrelation** across intervals and may be modelled as an autoregressive process or sampled from historical wind profiles.
- $\tilde{A}_{S,t}$ follows a known diurnal curve scaled by a stochastic weather factor, rendering it **conditionally deterministic** given cloud cover.
- Under the present specification, D_t is **independent of** $\tilde{A}_{W,t}$ and $\tilde{A}_{S,t}$ by assumption. In reality, common weather drivers induce correlation between wind availability and demand (e.g., high wind at night coincides with low demand); this is a known source of model error and a candidate for later refinement.
- Competitor bids may remain **correlated with demand** through common weather and load signals even if demand itself is treated as independent of renewable availability.

You can also make the following assumptions regarding the behavioural logic of the producers: (1) conventional producers have stable bid quantities, with bid prices following normal distributions with uncertain means and variances; (2) the solar-power producer's bid quantity follows $\max(0, a + b \cos(\frac{2\pi t}{24}))$, with bid prices following a normal distribution with uncertain mean and variance; and (3) actual wind-power production is stochastic, following a normal distribution with constant mean and variance. Please refine your specification accordingly and present the revised section.

Revised Sections: Bid Formation, Constants, and Stochastic Variables

The new behavioural assumptions have material consequences for Sections 6.2, 8, and 9. Each is restated in full below.

6.2 Bid Formation (Revised)

Behavioural logic now differs by producer type.

Conventional Producers $i \in \mathcal{C}$

Bid quantities are stable and set equal to each producer's installed capacity:

$$q_{i,t}^{\text{bid}} = \bar{Q}_i, \quad \forall t \in \mathcal{T}$$

Bid prices are drawn independently each interval from a normal distribution:

$$p_{i,t}^{\text{bid}} \sim \mathcal{N}(\mu_i^p, \sigma_i^{p^2}), \quad \forall t \in \mathcal{T}$$

where μ_i^p and $\sigma_i^{p^2}$ are uncertain parameters reflecting the unknown short-run marginal cost structure of producer i (driven by fuel prices, plant efficiency, and operating conditions). As with demand, realisations are truncated below at zero to preserve physical admissibility: $p_{i,t}^{\text{bid}} = \max(0, \mathcal{N}(\mu_i^p, \sigma_i^{p^2}))$.

Solar Producer $i = S$

The bid quantity follows a deterministic diurnal profile governed by a cosine function of the hour index t :

$$q_{S,t}^{\text{bid}} = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right), \quad \forall t \in \mathcal{T}$$

where a is a baseline offset and b is an amplitude parameter. The $\max(0, \cdot)$ operator ensures non-negative quantities during intervals of negligible irradiance. Both a and b are constants of the model, calibrated from historical solar output data for the installation.

The bid price is drawn independently each interval from a normal distribution:

$$p_{S,t}^{\text{bid}} \sim \mathcal{N}(\mu_S^p, \sigma_S^{p^2}), \quad \forall t \in \mathcal{T}$$

where μ_S^p and $\sigma_S^{p^2}$ are uncertain parameters. Since the solar producer's marginal cost is near zero, μ_S^p is expected to be small, but uncertainty in bidding strategy justifies a distributional treatment.

Wind Producer $i = W$

Actual wind production in each interval is stochastic, drawn independently from a normal distribution with constant mean and variance:

$$\tilde{A}_{W,t} \sim \mathcal{N}(\mu_W, \sigma_W^2), \quad \forall t \in \mathcal{T}$$

This replaces the autocorrelated process assumed in the prior specification. The wind producer's bid quantity is physically constrained to lie within available production:

$$q_{W,t}^{\text{bid}} \leq \max(0, \tilde{A}_{W,t}), \quad \forall t \in \mathcal{T}$$

The bid price $p_{W,t}^{\text{bid}}$ remains a control variable chosen by the wind producer. Since the wind producer's marginal cost is near zero, a natural baseline strategy is $p_{W,t}^{\text{bid}} = 0$ to ensure dispatch whenever the bid is technically feasible; alternative pricing strategies may be examined in scenario analyses.

8. Constants (Revised)

Symbol	Description
$\bar{Q}_i$	Installed capacity of conventional producer $i \in \mathcal{C}$ (MW); also the fixed bid quantity
$\bar{Q}_W$	Nameplate capacity of the wind producer (MW)
$\bar{Q}_S$	Nameplate capacity of the solar producer (MW)
μ_D	Mean hourly electricity demand (MWh)
σ_D^2	Variance of hourly electricity demand (MWh ²)
μ_W	Mean available wind production per interval (MWh)
σ_W^2	Variance of available wind production per interval (MWh ²)
a	Baseline offset parameter of the solar bid-quantity profile (MWh)
b	Amplitude parameter of the solar bid-quantity profile (MWh)

Symbol	Description
$n_C = 3$	Number of conventional producers
$ \mathcal{T} = 24$	Number of trading intervals per day

9. Uncertain Parameters and Stochastic Variables (*Revised*)

9.1 Uncertain Parameters

These quantities are fixed for a given day but unknown to the modeller at bid submission time. They enter the model as parameters whose true values are not observed and must either be estimated from data or treated as random variables at the scenario level.

Symbol	Description
$\mu_i^p, \sigma_i^{p^2}$	Mean and variance of the bid price distribution for conventional producer $i \in \mathcal{C}$ (\\$/MWh); reflects unobserved fuel costs and operational conditions
$\mu_S^p, \sigma_S^{p^2}$	Mean and variance of the bid price distribution for the solar producer (\\$/MWh)

9.2 Stochastic Variables

Symbol	Distribution	Description
D_t	$\mathcal{N}(\mu_D, \sigma_D^2)$	Electricity demand in interval t (MWh)
$p_{i,t}^{\text{bid}}$	$\mathcal{N}(\mu_i^p, \sigma_i^{p^2})$	Bid price of conventional producer i in interval t (\\$/MWh)
$p_{S,t}^{\text{bid}}$	$\mathcal{N}(\mu_S^p, \sigma_S^{p^2})$	Bid price of solar producer in interval t (\\$/MWh)
$\tilde{A}_{W,t}$	$\mathcal{N}(\mu_W, \sigma_W^2)$	Available wind production in interval t (MWh)

9.3 Distributional Assumptions and Caveats

Independence across intervals. All four stochastic variables are assumed independent across intervals t . For wind production, this supersedes the autocorrelated process in the prior specification. Serial correlation in wind speed is a well-documented empirical regularity; the i.i.d. assumption is adopted here for tractability and constitutes a known simplification.

Independence across variables. Demand D_t , wind availability $\tilde{A}_{W,t}$, and the competitor bid prices are treated as mutually independent within each interval. In practice, common weather drivers induce correlation between wind output and demand, and between weather conditions and competitor cost structures. This is a candidate for refinement in later model extensions.

Support truncation. All four variables are normally distributed and may in principle take negative values, which is physically inadmissible in each case. The truncation condition $\mu \gg \sigma$ is assumed to hold for each distribution, rendering the probability of a non-positive realisation negligible. Where this cannot be assured — particularly for wind production, which may be genuinely near zero for extended periods — a truncated normal $\mathcal{N}^+(\mu, \sigma^2)$ restricted to $\mathbb{R}_{\geq 0}$ should be substituted.

Solar bid quantity. The solar bid quantity $q_{S,t}^{\text{bid}} = \max(0, a + b \cos(2\pi t/24))$ is fully deterministic given the constants a and b . It is therefore not a stochastic variable; any uncertainty in solar output is captured indirectly through the uncertain bid price parameters μ_S^p and $\sigma_S^{p^2}$. If stochastic irradiance variation is deemed material, a random perturbation term $\epsilon_{S,t}$ may be added to the profile in a subsequent extension.

When multiple bids at the clearing price collectively exceed the residual demand, producers are dispatched in priority order: solar $\text{\$succ\$}$ wind $\text{\$succ\$}$ conv1 $\text{\$succ\$}$ conv2 $\text{\$succ\$}$ conv3, with the last accepted producer potentially dispatched only partially to exactly meet demand. Please refine your specification accordingly and present the revised section.

6.3 Merit-Order Market Clearing (Revised)

For each interval t , the market operator executes the following procedure.

Step 1 — Construct the Supply Stack

Let $\mathcal{I} = \{S, W, 1, 2, 3\}$ denote the full set of producers. Sort all bids in non-decreasing order of bid price to form the merit-order stack. Where two or more producers submit identical bid prices, they are ordered within that price tier according to the priority rule:

$$S \succ W \succ \text{conv}_1 \succ \text{conv}_2 \succ \text{conv}_3$$

Let $\pi_t = (\pi_t(1), \pi_t(2), \pi_t(3), \pi_t(4), \pi_t(5))$ denote the resulting ordered sequence of producers for interval t , combining price-based and priority-based ordering.

Step 2 — Identify the Marginal Producer

Traverse the stack in order, accumulating dispatched quantity until demand D_t is met. Define the cumulative quantity after accepting the first k producers in the stack:

$$\mathcal{Q}_t(k) = \sum_{j=1}^k q_{\pi_t(j), t}^{\text{bid}}$$

The marginal producer rank m_t is the index of the first producer whose inclusion causes cumulative supply to meet or exceed demand:

$$m_t = \min \{k \in \{1, \dots, 5\} \mid \mathcal{Q}_t(k) \geq D_t\}$$

The market-clearing price is set to the marginal producer's bid price:

$$\lambda_t = p_{\pi_t(m_t), t}^{\text{bid}}$$

Step 3 — Determine Dispatch Quantities

All producers ranked strictly above the marginal producer are accepted in full. The marginal producer is dispatched only to the extent required to exactly satisfy residual demand. All producers ranked below the marginal producer are rejected.

The residual demand facing the marginal producer is:

$$D_t^{\text{res}} = D_t - \mathcal{Q}_t(m_t - 1)$$

where $\mathcal{Q}_t(0) \equiv 0$. Dispatched quantities are then:

$$q_{\pi_t(j), t}^{\text{disp}} = \begin{cases} q_{\pi_t(j), t}^{\text{bid}} & \text{if } j < m_t \\ D_t^{\text{res}} & \text{if } j = m_t \\ 0 & \text{if } j > m_t \end{cases}$$

Note that $0 < D_t^{\text{res}} \leq q_{\pi_t(m_t), t}^{\text{bid}}$ by construction, so the marginal producer is always at least partially dispatched and never over-dispatched.

Step 4 — Revenue Calculation

Each accepted producer is remunerated at the clearing price for the quantity actually dispatched:

$$r_{i,t} = q_{i,t}^{\text{disp}} \cdot \lambda_t, \quad \forall i \in \mathcal{I}$$

The wind producer's total daily revenue is then:

$$R_W = \sum_{t=1}^{24} r_{W,t} = \sum_{t=1}^{24} q_{W,t}^{\text{disp}} \cdot \lambda_t$$

Remarks

Role of the priority rule. The priority ordering $S \succ W \succ \text{conv}_1 \succ \text{conv}_2 \succ \text{conv}_3$ only becomes operative when two or more producers submit identical bid prices — most consequentially when multiple producers bid at or near zero, which is the natural strategy for both the solar and wind producers given their near-zero marginal costs. In such cases, solar is guaranteed full dispatch before wind is considered, and wind before any conventional producer, which has a direct bearing on the wind producer's expected dispatched quantity and hence its revenue.

****Partial dispatch and revenue implications.**** When the wind producer is the marginal producer ($\pi_t(m_t) = W$), it is dispatched at $D_t^{\text{res}} \leq q_{W,t}^{\text{bid}}$, earning less than it would under full dispatch. This outcome is more likely when total low-cost supply (solar plus wind) approaches or exceeds demand, a condition whose probability depends on the joint distribution of $\tilde{A}_{W,t}$, $q_{S,t}^{\text{bid}}$, and D_t .

Market feasibility. The clearing procedure implicitly requires that total offered supply is sufficient to meet demand in every interval:

$$\sum_{i \in \mathcal{I}} q_{i,t}^{\text{bid}} \geq D_t, \quad \forall t \in \mathcal{T}$$

Given that demand and wind availability are both stochastic, this condition is not guaranteed to hold in every simulation draw. If it is violated, the market cannot be cleared by the merit-order mechanism alone. A supply-shortfall handling rule — such as involuntary load shedding at a fixed value of lost load, or rejection of the draw — should be specified before simulation commences.

Given that actual wind-power production is stochastic, it may fall short of the quantity committed to dispatch one day in advance. If the wind-power producer under-delivers relative to its dispatched quantity, the shortfall incurs a constant penalty cost (\$/MWh). Please incorporate this penalty mechanism into your specification.

Then, please provide a complete model specification that specifies all key components, including the time scales, agents, state variables, control variables, processes, objectives, constants, uncertain parameters and stochastic variables.

Complete Agent-Based Model Specification

Day-Ahead Wind-Power Revenue Estimation

1. Overview

The model simulates a single day of operations in a day-ahead electricity spot market. Five producer agents submit price-quantity bids for 24 hourly intervals before the target day begins. A market operator agent clears each interval via a merit-order mechanism with a defined priority rule for tied bids. After market clearing, actual wind production is realised stochastically; any shortfall relative to the dispatched quantity incurs a penalty cost. The primary output is the wind producer's total daily net revenue, which constitutes one sample from the distribution of realisable revenues under uncertainty.

2. Time Scales

Scale	Description
Simulation horizon	One target day, indexed d
Trading intervals	24 hourly slots, indexed $t \in \mathcal{T} = \{1, 2, \dots, 24\}$

Scale	Description
Bid submission	A single event occurring on day $d - 1$; all bids for all 24 intervals are submitted simultaneously, prior to any realisation of stochastic variables
Market clearing	One clearing event per interval t , executed sequentially after bid submission; determines λ_t and $q_{i,t}^{\text{disp}}$ for all i
Physical delivery	Occurs in real time during interval t on day d ; actual wind production $\tilde{A}_{W,t}$ is realised and compared against the committed dispatch quantity

The separation between bid submission and physical delivery is the structural source of delivery risk for the wind producer: bids are placed under uncertainty about actual wind output, and penalties are assessed ex post against commitments made ex ante.

3. Agents

3.1 Producer Agents

There are five producer agents, $\mathcal{I} = \{S, W, 1, 2, 3\}$, partitioned by technology type.

Conventional Producers $i \in \mathcal{C} = \{1, 2, 3\}$

Dispatchable coal-fired generators. Bid quantities are set equal to installed capacity. Bid prices reflect uncertain short-run marginal costs, drawn from normal distributions. These agents are price-takers with no strategic behaviour modelled at this stage.

Solar Producer $i = S$

A non-dispatchable generator whose bid quantity follows a deterministic diurnal cosine profile. Its marginal cost is near zero. Its bid price is drawn from a normal distribution with uncertain mean and variance. It faces no delivery penalty as its bid quantity profile is treated as a firm commitment matched by a deterministic output schedule.

Wind Producer $i = W$

The focal agent. Its bid quantity and price are the primary control variables. Actual production is stochastic and unknown at bid submission time; if realised output falls short of the dispatched quantity, a per-MWh penalty is incurred. The wind producer's objective is to choose bids that maximise expected net revenue, trading off the risk of under-delivery against the benefit of higher dispatch probability.

3.2 Market Operator Agent

A single non-strategic agent that collects all bids, applies the merit-order clearing mechanism with priority-based tie-breaking for each interval, and publishes the clearing price and dispatch schedule. It has no objective function and executes its rule deterministically.

4. State Variables

Symbol	Domain	Description
$q_{i,t}^{\text{bid}}$	$\mathbb{R}_{\geq 0}$	Quantity bid by producer i for interval t (MWh)
$p_{i,t}^{\text{bid}}$	$\mathbb{R}_{\geq 0}$	Minimum price bid by producer i for interval t ($\$/\text{MWh}$)
π_t	Ordered sequence	Merit-order stack for interval t , combining price ranking and priority-based tie-breaking
m_t	$\{1, \dots, 5\}$	Rank of the marginal producer in stack π_t
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price for interval t ($\$/\text{MWh}$)
$q_{i,t}^{\text{disp}}$	$\mathbb{R}_{\geq 0}$	Quantity dispatched from producer i at interval t (MWh)
$\tilde{A}_{W,t}$	$\mathbb{R}$	Actual wind production realised during interval t (MWh); drawn after market clearing
$s_{W,t}$	$\mathbb{R}_{\geq 0}$	Delivery shortfall of the wind producer in interval t (MWh); $s_{W,t} = (q_{W,t}^{\text{disp}} - \tilde{A}_{W,t})^+$

Symbol	Domain	Description
$r_{W,t}^{\text{net}}$	$\mathbb{R}$	Net revenue of the wind producer in interval t (\\$)
R_W^{net}	$\mathbb{R}$	Total daily net revenue of the wind producer (\\$)

5. Control Variables

The wind producer selects the following decision variables at bid submission time, prior to observing any stochastic realisations:

Symbol	Feasible Set	Description
$q_{W,t}^{\text{bid}}$	$[0, \bar{Q}_W]$	Quantity the wind producer offers to supply in interval t (MWh); bounded by nameplate capacity
$p_{W,t}^{\text{bid}}$	$[0, \infty)$	Minimum price the wind producer is willing to accept in interval t (\\$/MWh)

The bid quantity $q_{W,t}^{\text{bid}}$ is no longer physically constrained by $\tilde{A}_{W,t}$ at the time of bidding, since actual production has not yet been realised. Instead, the penalty mechanism disciplines over-bidding economically: bidding more than expected production increases dispatch probability but raises expected penalty exposure. The optimal bid therefore reflects a balance between these competing effects.

Bids of all other producers are treated as exogenous draws from their respective distributions and are not decision variables in the present model.

6. Processes

6.1 Exogenous Variable Realisation (Day $d - 1$, Pre-Bid)

The following quantities, though not directly observable by the wind producer at bid time, govern the simulation draw for the target day. Demand and competitor bids are realised per interval during clearing; wind production is realised during delivery.

$$D_t \sim \mathcal{N}(\mu_D, \sigma_D^2), \quad \forall t \in \mathcal{T}$$

$$p_{i,t}^{\text{bid}} \sim \mathcal{N}(\mu_i^p, \sigma_i^{p^2}), \quad \forall i \in \mathcal{C}, t \in \mathcal{T}$$

$$p_{S,t}^{\text{bid}} \sim \mathcal{N}(\mu_S^p, \sigma_S^{p^2}), \quad \forall t \in \mathcal{T}$$

$$\tilde{A}_{W,t} \sim \mathcal{N}(\mu_W, \sigma_W^2), \quad \forall t \in \mathcal{T}$$

All realisations are treated as independent across intervals and across variables.

6.2 Bid Formation (Day $d - 1$, Submission)

****Conventional producers**** bid their full installed capacity at a drawn price:

$$q_{i,t}^{\text{bid}} = \bar{Q}_i, \quad p_{i,t}^{\text{bid}} = \max(0, \mathcal{N}(\mu_i^p, \sigma_i^{p^2})), \quad \forall i \in \mathcal{C}$$

Solar producer bids a deterministic diurnal quantity at a drawn price:

$$q_{S,t}^{\text{bid}} = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right), \quad p_{S,t}^{\text{bid}} = \max(0, \mathcal{N}(\mu_S^p, \sigma_S^{p^2}))$$

Wind producer submits control variables ($q_{W,t}^{\text{bid}}, p_{W,t}^{\text{bid}}$) chosen to maximise expected net revenue (see Section 7), subject to $0 \leq q_{W,t}^{\text{bid}} \leq \bar{Q}_W$.

6.3 Merit-Order Market Clearing (Day $d - 1$, Post-Submission)

For each interval $t \in \mathcal{T}$, the market operator executes the following.

Step 1 — Construct the supply stack. Sort all bids in non-decreasing order of bid price. Within any price tier where two or more producers submit identical prices, apply the

priority ordering:

$$S \succ W \succ \text{conv}_1 \succ \text{conv}_2 \succ \text{conv}_3$$

Let $\pi_t = (\pi_t(1), \dots, \pi_t(5))$ denote the resulting ordered sequence.

Step 2 — Identify the marginal producer. Define cumulative supply after the first k producers in the stack:

$$\mathcal{Q}_t(k) = \sum_{j=1}^k q_{\pi_t(j),t}^{\text{bid}}, \quad \mathcal{Q}_t(0) \equiv 0$$

The marginal rank is:

$$m_t = \min \{k \in \{1, \dots, 5\} \mid \mathcal{Q}_t(k) \geq D_t\}$$

The market-clearing price is:

$$\lambda_t = p_{\pi_t(m_t),t}^{\text{bid}}$$

Step 3 — Determine dispatch quantities. The residual demand facing the marginal producer is $D_t^{\text{res}} = D_t - \mathcal{Q}_t(m_t - 1)$. Dispatched quantities are:

$$q_{\pi_t(j),t}^{\text{disp}} = \begin{cases} q_{\pi_t(j),t}^{\text{bid}} & \text{if } j < m_t \\ D_t^{\text{res}} & \text{if } j = m_t \\ 0 & \text{if } j > m_t \end{cases}$$

By construction, $0 < D_t^{\text{res}} \leq q_{\pi_t(m_t),t}^{\text{bid}}$, so the marginal producer is always at least partially dispatched and never over-dispatched.

Market feasibility condition. Clearing requires $\mathcal{Q}_t(5) \geq D_t$. Since demand and some supply quantities are stochastic, this is not guaranteed. If violated, a supply-shortfall rule must be invoked — for example, curtailing demand at a pre-specified value of lost load, or discarding and resampling the draw.

6.4 Physical Delivery and Penalty Assessment (Day d , Interval t)

After market clearing, actual wind production $\tilde{A}_{W,t}$ is realised. The delivery shortfall is:

$$s_{W,t} = \left(q_{W,t}^{\text{disp}} - \tilde{A}_{W,t} \right)^+$$

If $\tilde{A}_{W,t} \geq q_{W,t}^{\text{disp}}$, the wind producer meets its obligation in full and no penalty is incurred; any surplus production above the dispatched quantity is curtailed and yields no additional revenue. The wind producer's net revenue for interval t is:

$$\begin{aligned} r_{W,t}^{\text{net}} &= q_{W,t}^{\text{disp}} \cdot \lambda_t - \kappa \cdot s_{W,t} \\ &= q_{W,t}^{\text{disp}} \cdot \lambda_t - \kappa \cdot \left(q_{W,t}^{\text{disp}} - \tilde{A}_{W,t} \right)^+ \end{aligned}$$

The penalty term $\kappa \cdot s_{W,t}$ represents the cost of procuring replacement energy or settling the imbalance in the balancing market at the penalty rate κ .

6.5 Daily Revenue Aggregation

The wind producer's total daily net revenue is:

$$R_W^{\text{net}} = \sum_{t=1}^{24} r_{W,t}^{\text{net}} = \sum_{t=1}^{24} \left[q_{W,t}^{\text{disp}} \cdot \lambda_t - \kappa \cdot \left(q_{W,t}^{\text{disp}} - \tilde{A}_{W,t} \right)^+ \right]$$

7. Objectives

Agent	Objective
Wind producer	$\max_{\{q_{W,t}^{\text{bid}}, p_{W,t}^{\text{bid}}\}_{t=1}^{24}} \mathbb{E}[R_W^{\text{net}}]$, subject to $q_{W,t}^{\text{bid}} \in [0, \bar{Q}_W]$
Solar producer	Maximise expected daily revenue; bid price set near zero to ensure dispatch
Conventional producers	Maximise profit; bid price set to recover uncertain marginal cost
Market operator	No optimisation objective; executes merit-order rule deterministically
Modeller	Estimate the distribution of R_W^{net} across simulation runs to assess the economic viability of the wind investment

The wind producer's optimisation problem is non-trivial: a higher bid quantity $q_{W,t}^{\text{bid}}$ increases the probability of full dispatch and, conditional on dispatch, the gross revenue $q_{W,t}^{\text{disp}} \cdot \lambda_t$, but simultaneously raises expected penalty exposure $\kappa \cdot \mathbb{E}[s_{W,t}]$. A lower bid quantity reduces penalty risk but foregoes dispatch revenue. The optimal bid balances these forces and will generally depend on $\mu_W, \sigma_W^2, \kappa$, and the anticipated clearing price distribution.

8. Constants

Symbol	Description
$\bar{Q}_i$	Installed capacity of conventional producer $i \in \mathcal{C}$ (MW); equals fixed bid quantity
$\bar{Q}_W$	Nameplate capacity of the wind producer (MW); upper bound on $q_{W,t}^{\text{bid}}$
$\bar{Q}_S$	Nameplate capacity of the solar producer (MW)
μ_D, σ_D^2	Mean (MWh) and variance (MWh ²) of hourly electricity demand
μ_W, σ_W^2	Mean (MWh) and variance (MWh ²) of actual wind production per interval
a, b	Baseline offset (MWh) and amplitude (MWh) of the solar bid-quantity cosine profile
κ	Penalty rate for wind under-delivery (\$/MWh); constant across intervals
$n_C = 3$	Number of conventional producers
$ \mathcal{T} = 24$	Number of trading intervals per day

9. Uncertain Parameters and Stochastic Variables

9.1 Uncertain Parameters

Fixed for a given simulation scenario but not directly observed; must be estimated from data or subjected to sensitivity analysis.

Symbol	Description
μ_i^p, σ_i^{p2}	Mean (\$/MWh) and variance of the bid price distribution for conventional producer $i \in \mathcal{C}$
μ_S^p, σ_S^{p2}	Mean (\$/MWh) and variance of the bid price distribution for the solar producer

9.2 Stochastic Variables

Symbol	Distribution	Timing	Description
D_t	$\mathcal{N}(\mu_D, \sigma_D^2)$	Clearing	Electricity demand in interval t (MWh)
$p_{i,t}^{\text{bid}}$	$\mathcal{N}(\mu_i^p, \sigma_i^{p2})$	Bid submission	Bid price of conventional producer i in interval t (\$/MWh)
$p_{S,t}^{\text{bid}}$	$\mathcal{N}(\mu_S^p, \sigma_S^{p2})$	Bid submission	Bid price of solar producer in interval t (\$/MWh)

Symbol	Distribution	Timing	Description
$\tilde{A}_{W,t}$	$\mathcal{N}(\mu_W, \sigma_W^2)$	Delivery	Actual wind production in interval t (MWh); realised after market clearing

9.3 Distributional Assumptions and Caveats

Independence across intervals. All stochastic variables are assumed i.i.d. across t . Serial autocorrelation in wind production and demand — a well-documented empirical regularity — is abstracted away for tractability and represents a known source of model error.

Independence across variables. Demand, wind production, and competitor bid prices are treated as mutually independent within each interval. In practice, common weather drivers induce cross-variable correlation; this is a candidate for refinement in later extensions.

Support truncation. All variables are normally distributed and may in principle take physically inadmissible negative values. The condition $\mu \gg \sigma$ is assumed to hold for each distribution, rendering such realisations negligible in probability. Where this cannot be assured — particularly for wind production under high variance — a truncated normal $\mathcal{N}^+(\mu, \sigma^2)$ restricted to $\mathbb{R}_{\geq 0}$ should be substituted.

****Timing of wind realisation.**** The variable $\tilde{A}_{W,t}$ is drawn after market clearing, reflecting the physical reality that actual wind output on the target day is unknown at bid submission. This timing is structurally central to the penalty mechanism: the shortfall $s_{W,t} = (q_{W,t}^{\text{disp}} - \tilde{A}_{W,t})^+$ can only be evaluated ex post.

Solar bid quantity. The quantity $q_{S,t}^{\text{bid}} = \max(0, a + b \cos(2\pi t/24))$ is fully deterministic given the constants a and b and carries no stochastic component at this stage. If variable cloud cover is deemed material, a random perturbation $\epsilon_{S,t}$ may be added to the profile in a subsequent extension.

10. Summary Schematic

DAY d-1: BID SUBMISSION

```

Conv i:   q_bid = Q_i   | p_bid ~ N(μ_i^p, σ_i^p^2)
Solar:    q_bid = max(0, a+b*cos(2πt/24)) | p_bid ~ N(μ_S^p, σ_S^p^2)
Wind:     q_bid ∈ [0, Q_W] | p_bid ← CONTROL VARIABLES

```

↓

DAY d-1: MARKET CLEARING (repeated for each $t = 1, \dots, 24$)

```

Draw D_t ~ N(μ_D, σ_D^2)
Sort bids → merit-order stack π_t
(ties broken by: S > W > conv1 > conv2 > conv3)
Identify marginal producer m_t → set λ_t
Dispatch: full for j < m_t | partial for j = m_t | zero for j > m_t

```

↓

DAY d: PHYSICAL DELIVERY (interval t)

```

Draw  $\tilde{A}_{\{W,t\}} \sim \mathcal{N}(\mu_W, \sigma_W^2)$ 
Shortfall:  $s_{\{W,t\}} = \max(0, q_{\text{disp}_{\{W,t\}}} - \tilde{A}_{\{W,t\}})$ 
Net revenue:  $r_{\{W,t\}} = q_{\text{disp}_{\{W,t\}}} \cdot \lambda_t - \kappa \cdot s_{\{W,t\}}$ 

```

↓

DAILY AGGREGATION

```

R_W^net = Σ_{t=1}^{24} r_{\{W,t\}}^net

```

11. Model Extensions (Deferred)

The following components are noted for future development but excluded from the single-day model:

- **Multi-day simulation** with learning, adaptive bidding, and accumulating revenue distributions
- **Seasonal and inter-annual resource variability** for long-run investment viability assessment
- **Strategic bidding** by competitors modelled as a game-theoretic equilibrium problem

- **Temporal dependence** in wind production and demand via autoregressive or regime-switching processes
- **Cross-variable correlation** between wind availability, demand, and competitor costs via a joint distributional model
- **Transmission constraints** and locational marginal pricing
- **Capacity payments** and ancillary service revenue streams
- **Carbon pricing** effects on conventional producer marginal costs

Various stakeholders approach this problem from different perspectives. Please revise your specification to use agents to represent two perspectives separately and independently: (1) the wind-power producer; and (2) the system regulator. In the revised specification, specify the common environment, including the agents, global constants, uncertain parameters and stochastic variables. For each perspective, specify the agent's key components, including state variables, control variables, processes and objectives.

Agent-Based Model Specification: Dual-Perspective Framework

Day-Ahead Wind-Power Revenue Estimation

Part I: Common Environment

1. Overview

The model simulates a single day of operations in a day-ahead electricity spot market. The environment is shared by all agents and defines the market structure, clearing mechanism, and sources of uncertainty that are common to both perspectives. Two agents — the **wind-power producer** and the **system regulator** — interact with this environment independently, each with distinct state variables, control variables, processes, and objectives. The common environment is not owned by either agent; it constitutes the objective reality within which both operate.

2. Market Structure and Time Scales

Scale	Description
Simulation horizon	One target day, indexed d
Trading intervals	24 hourly slots, indexed $t \in \mathcal{T} = \{1, 2, \dots, 24\}$
Bid submission	A single event on day $d - 1$; all producers submit bids for all 24 intervals simultaneously, prior to any realisation of stochastic variables
Market clearing	One clearing event per interval t , executed sequentially after bid submission
Physical delivery	Real-time delivery during interval t on day d ; actual wind production is realised and compared against committed dispatch quantities

3. Agents

Five producer agents and one market operator agent populate the environment.

Agent	Type	Role
$i \in \mathcal{C} = \{1, 2, 3\}$	Conventional (coal)	Submit fixed-quantity, stochastic-price bids; fully dispatchable
$i = \mathcal{S}$	Solar	Submit diurnal-profile quantity bids at near-zero stochastic prices

Agent	Type	Role
$i = W$	Wind	Focal agent; submits bids under output uncertainty; subject to delivery penalty
Market operator	Mechanism	Clears each interval via merit-order with priority-based tie-breaking; non-strategic

The **wind-power producer** and **system regulator** are the two perspective agents whose specifications are developed in Parts II and III respectively. They do not appear as separate actors in the market mechanism but represent two analytical lenses applied to the same simulated system.

4. Market Clearing Mechanism

For each interval t , the market operator constructs the merit-order supply stack π_t by sorting bids in non-decreasing order of price. Ties are broken by the fixed priority rule:

$$S \succ W \succ \text{conv}_1 \succ \text{conv}_2 \succ \text{conv}_3$$

The marginal producer rank is:

$$m_t = \min \{k \in \{1, \dots, 5\} \mid \mathcal{Q}_t(k) \geq D_t\}, \quad \mathcal{Q}_t(k) = \sum_{j=1}^k q_{\pi_t(j), t}^{\text{bid}}$$

The clearing price is $\lambda_t = p_{\pi_t(m_t), t}^{\text{bid}}$. Dispatched quantities are:

$$q_{\pi_t(j), t}^{\text{disp}} = \begin{cases} q_{\pi_t(j), t}^{\text{bid}} & j < m_t \\ D_t - \mathcal{Q}_t(m_t - 1) & j = m_t \\ 0 & j > m_t \end{cases}$$

The feasibility condition $\mathcal{Q}_t(5) \geq D_t$ must hold for clearing to proceed. If violated, a supply-shortfall rule — such as load shedding at a pre-specified value of lost load, or rejection and resampling of the draw — must be invoked before simulation commences.

5. Global Constants

These quantities are fixed across all simulation runs and are shared by both agent perspectives.

Symbol	Description
$\bar{Q}_i$	Installed capacity of conventional producer $i \in \mathcal{C}$ (MW); equals fixed bid quantity
$\bar{Q}_W$	Nameplate capacity of the wind producer (MW)
$\bar{Q}_S$	Nameplate capacity of the solar producer (MW)
μ_D, σ_D^2	Mean (MWh) and variance (MWh ²) of hourly electricity demand
μ_W, σ_W^2	Mean (MWh) and variance (MWh ²) of actual wind production per interval
a, b	Baseline offset (MWh) and amplitude (MWh) of the solar bid-quantity cosine profile
κ	Penalty rate for wind under-delivery (\$/MWh)
$ \mathcal{T} = 24$	Number of trading intervals per day

6. Uncertain Parameters

Fixed for a given simulation scenario but not directly observed by any agent; must be estimated from data or subjected to sensitivity analysis.

Symbol	Description
$\mu_i^p, \sigma_i^{p^2}$	Mean (\$/MWh) and variance of the bid price distribution for conventional producer $i \in \mathcal{C}$

Symbol	Description
$\mu_S^p, \sigma_S^{p^2}$	Mean (\$/MWh) and variance of the bid price distribution for the solar producer

7. Stochastic Variables

Symbol	Distribution	Timing	Description
D_t	$\mathcal{N}(\mu_D, \sigma_D^2)$	Clearing	Electricity demand in interval t (MWh)
$p_{i,t}^{\text{bid}}$	$\mathcal{N}(\mu_i^p, \sigma_i^{p^2})$	Bid submission	Bid price of conventional producer i in interval t (\\$/MWh)
$p_{S,t}^{\text{bid}}$	$\mathcal{N}(\mu_S^p, \sigma_S^{p^2})$	Bid submission	Bid price of solar producer in interval t (\$/MWh)
$\tilde{A}_{W,t}$	$\mathcal{N}(\mu_W, \sigma_W^2)$	Delivery	Actual wind production realised in interval t (MWh); drawn after market clearing

All variables are assumed independent across intervals t and mutually independent within each interval. Negative realisations are physically inadmissible; the condition $\mu \gg \sigma$ is assumed to hold for each distribution. Where this cannot be assured, a truncated normal $\mathcal{N}^+(\mu, \sigma^2)$ restricted to $\mathbb{R}_{\geq 0}$ should be substituted.

Part II: Wind-Power Producer Agent

II.1 Role and Perspective

The wind-power producer is a price-taking generator operating under output uncertainty. It participates in the market by submitting bids for each interval of the target day before actual wind conditions are known, and is subsequently held financially accountable for any shortfall between its committed dispatch quantity and realised output. Its perspective is one of **decision-making under uncertainty**: given the stochastic environment, what bids maximise expected net revenue?

II.2 State Variables

Symbol	Domain	Description
$q_{W,t}^{\text{bid}}$	$[0, \bar{Q}_W]$	Bid quantity submitted for interval t (MWh)
$p_{W,t}^{\text{bid}}$	$\mathbb{R}_{\geq 0}$	Bid price submitted for interval t (\\$/MWh)
$q_{W,t}^{\text{disp}}$	$[0, q_{W,t}^{\text{bid}}]$	Quantity dispatched by the market operator in interval t (MWh)
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price observed in interval t (\\$/MWh)
$\tilde{A}_{W,t}$	$\mathbb{R}_{\geq 0}$	Actual wind production realised in interval t (MWh)
$s_{W,t}$	$\mathbb{R}_{\geq 0}$	Delivery shortfall in interval t (MWh); $s_{W,t} = (q_{W,t}^{\text{disp}} - \tilde{A}_{W,t})^+$
$r_{W,t}^{\text{net}}$	$\mathbb{R}$	Net revenue earned in interval t (\\$)
R_W^{net}	$\mathbb{R}$	Total daily net revenue (\$)

II.3 Control Variables

The wind producer selects these variables at bid submission time, before any stochastic realisation is observed.

Symbol	Feasible Set	Description
$q_{W,t}^{\text{bid}}$	$[0, \bar{Q}_W]$	Offered quantity for interval t (MWh)
$p_{W,t}^{\text{bid}}$	$[0, \infty)$	Minimum acceptable price for interval t (\\$/MWh)

The bid quantity is not physically constrained by $\tilde{A}_{W,t}$ at submission time, since actual output is unknown. The penalty mechanism disciplines over-bidding economically rather

than physically: higher bid quantities increase dispatch probability but raise expected penalty exposure. The optimal bid reflects a balance between these forces and will in general depend on $\mu_W, \sigma_W^2, \kappa$, and the anticipated clearing price distribution.

II.4 Processes

Bid formation. Prior to day d , the wind producer selects $(q_{W,t}^{\text{bid}}, p_{W,t}^{\text{bid}})$ for all $t \in \mathcal{T}$ to maximise expected net revenue, subject to the feasibility constraint $q_{W,t}^{\text{bid}} \in [0, \bar{Q}_W]$.

Dispatch observation. Following market clearing, the wind producer observes the clearing price λ_t and its dispatched quantity $q_{W,t}^{\text{disp}}$ for each interval.

Delivery and shortfall assessment. During interval t on day d , actual wind production $\tilde{A}_{W,t}$ is realised. The delivery shortfall is computed:

$$s_{W,t} = \left(q_{W,t}^{\text{disp}} - \tilde{A}_{W,t} \right)^+$$

If $\tilde{A}_{W,t} \geq q_{W,t}^{\text{disp}}$, the commitment is met in full; any surplus production above the dispatched quantity is curtailed and yields no additional revenue.

Net revenue calculation. The wind producer's net revenue for interval t is:

$$r_{W,t}^{\text{net}} = q_{W,t}^{\text{disp}} \cdot \lambda_t - \kappa \cdot s_{W,t}$$

Daily aggregation:

$$R_W^{\text{net}} = \sum_{t=1}^{24} r_{W,t}^{\text{net}} = \sum_{t=1}^{24} \left[q_{W,t}^{\text{disp}} \cdot \lambda_t - \kappa \cdot \left(q_{W,t}^{\text{disp}} - \tilde{A}_{W,t} \right)^+ \right]$$

II.5 Objective

$$\max_{\{q_{W,t}^{\text{bid}}, p_{W,t}^{\text{bid}}\}_{t=1}^{24}} \mathbb{E} [R_W^{\text{net}}]$$

subject to $q_{W,t}^{\text{bid}} \in [0, \bar{Q}_W]$ for all $t \in \mathcal{T}$.

The modeller's objective is to characterise the full distribution of R_W^{net} across simulation runs, providing summary statistics — expected value, variance, and tail quantiles — to support an assessment of the investment's economic viability.

Part III: System Regulator Agent

III.1 Role and Perspective

The system regulator is a non-market participant that observes the outcomes of market clearing and physical delivery across all producers and intervals. Its perspective is one of **system oversight**: given the collective behaviour of all market participants and the stochastic environment, how well does the market serve the goals of supply adequacy, price efficiency, and reliable integration of renewable generation? The regulator does not submit bids or earn revenue; instead, it monitors aggregate outcomes and, through its control over the penalty rate κ and market design parameters, shapes the incentive environment within which producers operate.

III.2 State Variables

Symbol	Domain	Description
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price in interval t ($\$/\text{MWh}$)
D_t	$\mathbb{R}_{> 0}$	Realised electricity demand in interval t (MWh)
$q_{i,t}^{\text{disp}}$	$\mathbb{R}_{\geq 0}$	Dispatched quantity of producer i in interval t (MWh)

Symbol	Domain	Description
$s_{W,t}$	$\mathbb{R}_{\geq 0}$	Wind delivery shortfall in interval t (MWh)
δ_t	$\{0, 1\}$	Supply adequacy indicator; $\delta_t = 1$ if $Q_t(5) \geq D_t$, else 0
ϕ_t	$[0, 1]$	Renewable share of dispatched generation in interval t ; $\phi_t = (q_{S,t}^{\text{disp}} + q_{W,t}^{\text{disp}})/D_t$
$\bar{\lambda}$	$\mathbb{R}_{\geq 0}$	Daily average clearing price ($\$/\text{MWh}$); $\bar{\lambda} = \frac{1}{24} \sum_{t=1}^{24} \lambda_t$
$\bar{\phi}$	$[0, 1]$	Daily average renewable penetration; $\bar{\phi} = \frac{1}{24} \sum_{t=1}^{24} \phi_t$
S_W^{total}	$\mathbb{R}_{\geq 0}$	Total daily wind delivery shortfall (MWh); $S_W^{\text{total}} = \sum_{t=1}^{24} s_{W,t}$

III.3 Control Variables

The regulator's control variables govern the incentive and institutional structure of the market. They are set prior to simulation and are treated as constants by all other agents, but are the instruments through which the regulator shapes market outcomes.

Symbol	Feasible Set	Description
κ	$[0, \infty)$	Penalty rate for wind under-delivery ($\$/\text{MWh}$); calibrated to balance delivery reliability against the wind producer's revenue risk
Π_{priority}	Ordered set over $\mathcal{I}$	Priority ordering for tie-breaking in the merit-order stack; the baseline is $S \succ W \succ \text{conv}_1 \succ \text{conv}_2 \succ \text{conv}_3$

III.4 Processes

Supply adequacy monitoring. For each interval t , the regulator checks whether the feasibility condition is satisfied:

$$\delta_t = \mathbf{1} \left[\sum_{i \in \mathcal{I}} q_{i,t}^{\text{bid}} \geq D_t \right]$$

Intervals where $\delta_t = 0$ constitute supply shortfall events, which are recorded and contribute to an adequacy performance metric across simulation runs.

Price formation monitoring. The regulator observes the sequence of clearing prices $\{\lambda_t\}_{t=1}^{24}$ and computes summary statistics including the daily mean $\bar{\lambda}$, the intra-day price range $\max_t \lambda_t - \min_t \lambda_t$, and the frequency of prices exceeding a regulatory threshold λ^{max} . These metrics indicate the degree of market efficiency and the presence of pricing anomalies.

Renewable integration assessment. For each interval t , the regulator computes the renewable penetration share ϕ_t and aggregates to the daily average $\bar{\phi}$. The regulator also records the joint distribution of (ϕ_t, λ_t) across intervals to assess whether higher renewable penetration is associated with lower or more volatile prices.

Delivery reliability assessment. The regulator observes the wind delivery shortfall $s_{W,t}$ for each interval and aggregates the total daily shortfall S_W^{total} . This quantity measures the extent to which committed wind dispatch was not physically delivered, with implications for system reliability. The regulator uses this information to evaluate whether the current penalty rate κ is sufficient to discipline over-bidding by the wind producer.

Penalty calibration. The regulator evaluates the adequacy of κ by assessing the trade-off between two observable outcomes across simulation runs: (i) the frequency and magnitude of wind delivery shortfalls, which increase as κ is set too low; and (ii) the wind producer's tendency to under-bid relative to expected output, which increases as κ is set too high and suppresses renewable dispatch. The regulator adjusts κ to balance these competing effects.

III.5 Objectives

The regulator pursues multiple objectives simultaneously, which may be in partial tension:

Objective	Formal Expression	Description
Supply adequacy	$\mathbb{E}\left[\sum_{t=1}^{24} \delta_t\right] = 24$	All intervals should be cleared without shortfall in expectation
Price efficiency	$\min \text{Var}(\lambda_t)$	Clearing prices should be stable and reflect true marginal costs
Renewable integration	$\mathbb{E}[\bar{\phi}] \geq \phi^{\min}$	Average renewable penetration should meet or exceed a policy target $\phi^{\min}$
Delivery reliability	$\mathbb{E}[S_W^{\text{total}}] \leq \bar{S}$	Expected total wind shortfall should not exceed a reliability tolerance $\bar{S}$
Penalty calibration	Find κ^*	Set the penalty rate to minimise shortfall frequency while preserving wind producer participation and dispatch incentives

The regulator's composite objective is to design a market environment — principally through the choice of κ and Π_{priority} — such that profit-seeking behaviour by the wind producer (Part II) produces system-level outcomes consistent with the objectives above. This defines a bilevel structure: the wind producer optimises its bids given κ , and the regulator chooses κ anticipating the wind producer's best response.

To ensure internal consistency across these perspectives, please compose them into a unified agent-based model. The composed model should: 1. embed each perspective's components, including state variables, control variables, processes and objectives; and 2. enable modular evaluation of each perspective, where other perspectives' controls are treated as exogenous inputs. Please specify the key components of the final model according to the ODD protocol.

Unified Agent-Based Model Specification

Day-Ahead Wind-Power Revenue Estimation

ODD Protocol

Section 1 — Purpose and Patterns

1.1 Purpose

The model simulates a single day of operations in a day-ahead electricity spot market comprising five generating producers and one market operator. Its primary purpose is to estimate the distribution of the wind-power producer's daily net revenue under uncertainty in wind output, electricity demand, and competitor behaviour. A secondary purpose is to evaluate the system-level consequences of market design choices — specifically the wind under-delivery penalty rate and the dispatch priority ordering — from the perspective of a system regulator. The model is intended to support an assessment of the economic viability of a wind-power investment and to characterise the sensitivity of that viability to regulatory parameters.

1.2 Patterns

The model is designed to reproduce and examine the following system-level patterns:

Pattern	Description
Revenue distribution	The distribution of R_W^{net} across simulation runs, including its mean, variance, and tail quantiles, as the primary investment viability indicator
Penalty-dispatch trade-off	The relationship between the penalty rate κ and the wind producer's optimal bid quantity, capturing the economic tension between over-delivery risk and foregone dispatch revenue

Pattern	Description
Merit-order price suppression	The tendency of low-cost renewable bids to depress clearing prices λ_t in intervals of high renewable availability
Renewable penetration profile	The intra-day pattern of renewable dispatch share ϕ_t as a function of the solar diurnal profile and stochastic wind availability
Supply adequacy events	The frequency of intervals in which aggregate bid quantities are insufficient to meet realised demand D_t

Section 2 — Entities, State Variables, and Scales

2.1 Scales

Dimension	Specification
Simulation horizon	One target day d
Trading intervals	$t \in \mathcal{T} = \{1, 2, \dots, 24\}$, each representing one hour
Bid submission	A single discrete event on day $d - 1$, prior to any stochastic realisation
Market clearing	One discrete event per interval t , executed sequentially after bid submission
Physical delivery	Continuous within interval t on day d ; modelled as a single end-of-interval realisation

2.2 Entities and State Variables

The model contains three entity types: **producer agents**, the **market operator agent**, and two **perspective agents**.

2.2.1 Conventional Producer Agents $i \in \mathcal{C} = \{1, 2, 3\}$

Variable	Domain	Description
$q_{i,t}^{\text{bid}}$	$\{\bar{Q}_i\}$	Bid quantity for interval t ; fixed at installed capacity (MWh)
$p_{i,t}^{\text{bid}}$	$\mathbb{R}_{\geq 0}$	Bid price for interval t ; drawn from $\mathcal{N}(\mu_i^p, \sigma_i^{p^2})$ ($\$/\text{MWh}$)
$q_{i,t}^{\text{disp}}$	$[0, \bar{Q}_i]$	Dispatched quantity in interval t (MWh)
$r_{i,t}$	$\mathbb{R}_{\geq 0}$	Revenue earned in interval t ($\$$)

2.2.2 Solar Producer Agent $i = S$

Variable	Domain	Description
$q_{S,t}^{\text{bid}}$	$\mathbb{R}_{\geq 0}$	Bid quantity for interval t ; deterministic diurnal profile (MWh)
$p_{S,t}^{\text{bid}}$	$\mathbb{R}_{\geq 0}$	Bid price for interval t ; drawn from $\mathcal{N}(\mu_S^p, \sigma_S^{p^2})$ ($\$/\text{MWh}$)
$q_{S,t}^{\text{disp}}$	$[0, q_{S,t}^{\text{bid}}]$	Dispatched quantity in interval t (MWh)
$r_{S,t}$	$\mathbb{R}_{\geq 0}$	Revenue earned in interval t ($\$$)

2.2.3 Wind Producer Agent $i = W$ — Perspective Agent 1

Variable	Domain	Description
$q_{W,t}^{\text{bid}}$	$[0, \bar{Q}_W]$	Control. Bid quantity for interval t (MWh)
$p_{W,t}^{\text{bid}}$	$[0, \infty)$	Control. Bid price for interval t ($\$/\text{MWh}$)
$q_{W,t}^{\text{disp}}$	$[0, q_{W,t}^{\text{bid}}]$	Dispatched quantity in interval t (MWh)
λ_t	$\mathbb{R}_{\geq 0}$	Observed clearing price in interval t ($\$/\text{MWh}$)

Variable	Domain	Description
$\tilde{A}_{W,t}$	$\mathbb{R}_{\geq 0}$	Actual wind production realised in interval t (MWh)
$s_{W,t}$	$\mathbb{R}_{\geq 0}$	Delivery shortfall in interval t ; $s_{W,t} = (q_{W,t}^{\text{disp}} - \tilde{A}_{W,t})^+$ (MWh)
$r_{W,t}^{\text{net}}$	$\mathbb{R}$	Net revenue in interval t (\\$)
R_W^{net}	$\mathbb{R}$	Total daily net revenue (\\$)

When evaluated in **wind-producer mode**, $(q_{W,t}^{\text{bid}}, p_{W,t}^{\text{bid}})$ are decision variables optimised by the agent. When evaluated in **regulator mode**, these are treated as exogenous inputs drawn from a specified bidding strategy.

2.2.4 Market Operator Agent

Variable	Domain	Description
π_t	Ordered sequence over $\mathcal{I}$	Merit-order stack for interval t , combining price ranking and priority tie-breaking
m_t	$\{1, \dots, 5\}$	Rank of the marginal producer in stack π_t
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price for interval t (\\$/MWh)
D_t^{res}	$\mathbb{R}_{> 0}$	Residual demand allocated to the marginal producer (MWh)
δ_t	$\{0, 1\}$	Feasibility indicator; $\delta_t = 1$ if aggregate supply meets demand

2.2.5 System Regulator Agent — Perspective Agent 2

Variable	Domain	Description
κ	$[0, \infty)$	Control. Penalty rate for wind under-delivery (\\$/MWh)
Π_{priority}	Ordered set over $\mathcal{I}$	Control. Priority ordering for merit-order tie-breaking
λ_t	$\mathbb{R}_{\geq 0}$	Observed clearing price in interval t (\\$/MWh)
ϕ_t	$[0, 1]$	Renewable share of dispatched generation in interval t
$\bar{\lambda}$	$\mathbb{R}_{\geq 0}$	Daily mean clearing price (\\$/MWh)
$\bar{\phi}$	$[0, 1]$	Daily mean renewable penetration
S_W^{total}	$\mathbb{R}_{\geq 0}$	Total daily wind delivery shortfall (MWh)
δ_t	$\{0, 1\}$	Observed supply adequacy indicator for interval t

When evaluated in **regulator mode**, $(\kappa, \Pi_{\text{priority}})$ are the decision variables being swept or optimised. When evaluated in **wind-producer mode**, these are fixed exogenous parameters received from the regulatory environment.

2.3 Modular Evaluation Structure

The two perspectives are composed such that each can be evaluated independently while the other's controls are held exogenous:

Evaluation Mode	Optimised Controls	Exogenous Inputs
Wind-producer mode	$q_{W,t}^{\text{bid}}, p_{W,t}^{\text{bid}}$	$\kappa, \Pi_{\text{priority}}$ fixed by regulator
Regulator mode	$\kappa, \Pi_{\text{priority}}$	$q_{W,t}^{\text{bid}}, p_{W,t}^{\text{bid}}$ supplied by a specified wind bidding strategy
Composed mode	Bilevel: regulator at upper level, wind producer at lower level	None; both perspectives active simultaneously

Section 3 — Process Overview and Scheduling

All processes execute in the following fixed sequence within each simulation run.

PHASE 0 – PARAMETER INITIALISATION

Set all global constants and uncertain parameters.

Regulator fixes $(\kappa, \Pi_{\text{priority}})$ – exogenous in wind-producer mode, decision variables in regulator mode.

PHASE 1 – BID SUBMISSION [Day d-1, single event]

For each $t \in T$:

Conv $i \in C$: draw $p_{\text{bid}}\{i, t\}$; set $q_{\text{bid}}\{i, t\} = Q_i$

Solar S : compute $q_{\text{bid}}\{S, t\}$ from cosine profile; draw $p_{\text{bid}}\{S, t\}$

Wind W : set $(q_{\text{bid}}\{W, t\}, p_{\text{bid}}\{W, t\})$ per active mode

PHASE 2 – MARKET CLEARING [Day d-1, repeated for each $t \in T$]

For each $t \in T$:

Draw $D_t \sim N(\mu_D, \sigma_D^2)$

Construct merit-order stack π_t using (Π_{priority})

Identify marginal producer m_t

Set clearing price λ_t

Compute dispatched quantities $q_{\text{disp}}\{i, t\}$ for all i

PHASE 3 – PHYSICAL DELIVERY [Day d, repeated for each $t \in T$]

For each $t \in T$:

Draw $\tilde{A}_{\{W, t\}} \sim N(\mu_W, \sigma_W^2)$

Compute shortfall $s_{\{W, t\}} = \max(0, q_{\text{disp}}\{W, t\} - \tilde{A}_{\{W, t\}})$

Compute net revenue $r_{\text{net}}\{W, t\} = q_{\text{disp}}\{W, t\} \cdot \lambda_t - \kappa \cdot s_{\{W, t\}}$

PHASE 4 – AGGREGATION AND OBSERVATION

Wind producer: $R_{\text{net}}^W = \sum_t r_{\text{net}}\{W, t\}$

Regulator: compute $\bar{\lambda}, \bar{\phi}, S_W^{\text{total}}, \{\delta_t\}$

Record all outputs for this simulation run

Section 4 — Design Concepts

4.1 Basic Principles

The model rests on three structural principles. First, the **merit-order clearing mechanism** reflects the standard economic logic of competitive electricity markets: producers are dispatched in ascending order of offered price until demand is satisfied, and all accepted producers are paid the marginal clearing price. Second, the **penalty mechanism** captures the asymmetric financial exposure of intermittent generators: a wind producer that over-commits relative to realised output bears an imbalance cost, while under-commitment merely foregoes revenue. Third, the **bilevel composition** of perspectives reflects the institutional reality that regulatory parameters $(\kappa, \Pi_{\text{priority}})$ are set by an authority external to the market, but their values shape the incentives that govern the wind producer's bidding behaviour.

4.2 Emergence

Market-clearing prices λ_t and the wind producer's net revenue R_W^{net} are emergent quantities: they arise from the interaction of five heterogeneous bidding agents through the merit-order mechanism and are not specified directly. The distribution of R_W^{net} across simulation runs is the primary emergent pattern of interest.

4.3 Adaptation

In wind-producer mode, the wind producer adapts its bid quantities and prices in response to the stochastic environment, the regulatory parameters $(\kappa, \Pi_{\text{priority}})$, and the anticipated behaviour of competing agents. The nature of this adaptation — whether it takes the form of analytical optimisation, heuristic rules, or scenario-based search — is a modelling choice to be specified at implementation. In the baseline formulation, the wind producer solves:

$$\max_{\{q_{W,t}^{\text{bid}}, p_{W,t}^{\text{bid}}\}_{t \in T}} \mathbb{E}[R_W^{\text{net}}]$$

In regulator mode, the regulator adapts κ to achieve system-level objectives, anticipating the wind producer's best-response bidding behaviour as a function of κ .

4.4 Objectives

Agent	Objective	Mode
Wind producer	$\max \mathbb{E}[R_W^{\text{net}}]$	Wind-producer mode
Regulator — supply adequacy	$\mathbb{E}[\sum_t \delta_t] = 24$	Regulator mode
Regulator — price efficiency	$\min \text{Var}(\lambda_t)$	Regulator mode
Regulator — renewable integration	$\mathbb{E}[\bar{\phi}] \geq \phi^{\min}$	Regulator mode
Regulator — delivery reliability	$\mathbb{E}[S_W^{\text{total}}] \leq \bar{S}$	Regulator mode
Regulator — penalty calibration	Find κ^* balancing shortfall and dispatch incentives	Regulator mode

4.5 Learning

No inter-run learning is implemented in the single-day model. Each simulation run is independent. In multi-day extensions, adaptive bidding strategies — such as reinforcement learning or Bayesian updating of competitor bid distributions — may be incorporated.

4.6 Prediction

The wind producer implicitly predicts the distribution of clearing prices λ_t when forming its bids. At this stage, these predictions are based on the known distributions of competitor bids and demand rather than on endogenous forecasting. The regulator implicitly predicts the wind producer's best-response bid as a function of κ when calibrating the penalty rate.

4.7 Sensing

Each producer agent observes its own bid and, after clearing, its own dispatch outcome and the clearing price. No agent observes the bids or dispatch outcomes of competitors prior to submission. The market operator observes all submitted bids simultaneously before clearing each interval. The regulator observes all post-clearing and post-delivery outcomes across all agents and intervals.

4.8 Interaction

Agent interactions are mediated entirely through the market mechanism: producers do not interact directly. The market operator aggregates bids into a supply stack and communicates dispatch quantities and the clearing price back to each producer. The regulator interacts with the market through its control variables ($\kappa, \Pi_{\text{priority}}$), which are set prior to simulation and affect all agents' incentive structures.

4.9 Stochasticity

Four independent stochastic processes drive model outcomes. Demand $D_t \sim \mathcal{N}(\mu_D, \sigma_D^2)$ and competitor bid prices $p_{i,t}^{\text{bid}} \sim \mathcal{N}(\mu_i^p, \sigma_i^{p^2})$ are drawn at clearing time. Actual wind production $\tilde{A}_{W,t} \sim \mathcal{N}(\mu_W, \sigma_W^2)$ is drawn at delivery time, after dispatch commitments have been made. All draws are independent across intervals and across variables. Negative realisations are physically inadmissible; the condition $\mu \gg \sigma$ is assumed to hold for all distributions, and truncated normals $\mathcal{N}^+(\mu, \sigma^2)$ should be substituted where this cannot be assured.

4.10 Collectives

No coalitions, portfolios, or aggregate collectives are modelled. Each producer acts independently.

4.11 Observation

The following quantities are recorded at the end of each simulation run:

Wind-producer perspective: $\{q_{W,t}^{\text{disp}}, \lambda_t, \tilde{A}_{W,t}, s_{W,t}, r_{W,t}^{\text{net}}\}_{t=1}^{24}$ and R_W^{net} .

Regulator perspective: $\{\lambda_t, \phi_t, \delta_t, s_{W,t}\}_{t=1}^{24}, \bar{\lambda}, \bar{\phi},$ and S_W^{total} .

Across N simulation runs, summary statistics — including sample means, variances, and empirical quantiles — are computed for all observed quantities to characterise the revenue

distribution and system performance metrics.

Section 5 — Initialisation

At the start of each simulation run, the following quantities are initialised.

Global constants (fixed across all runs):

Symbol	Description
$\bar{Q}_i, i \in \mathcal{C}$	Installed capacities of conventional producers (MW)
$\bar{Q}_W, \bar{Q}_S$	Nameplate capacities of wind and solar producers (MW)
μ_D, σ_D^2	Mean and variance of hourly electricity demand (MWh, MWh ²)
μ_W, σ_W^2	Mean and variance of actual wind production per interval (MWh, MWh ²)
a, b	Baseline offset and amplitude of the solar bid-quantity cosine profile (MWh)
κ	Wind under-delivery penalty rate (\$/MWh); fixed in wind-producer mode, variable in regulator mode
Π_{priority}	Priority ordering $S \succ W \succ \text{conv}_1 \succ \text{conv}_2 \succ \text{conv}_3$; fixed in wind-producer mode, variable in regulator mode

Uncertain parameters (fixed per run, estimated from data or subjected to sensitivity analysis):

Symbol	Description
μ_i^p, σ_i^{p2}	Mean and variance of bid price distribution for conventional producer $i \in \mathcal{C}$ (\$/MWh)
μ_S^p, σ_S^{p2}	Mean and variance of bid price distribution for solar producer (\$/MWh)

Agent state variables are reset to null at the start of each run and populated during simulation execution according to the process schedule in Section 3.

Section 6 — Input Data

The model does not read time-series input data at runtime in its baseline formulation. All stochastic variables are generated synthetically from their specified distributions using pseudorandom number generation. The distributional parameters ($\mu_D, \sigma_D^2, \mu_W, \sigma_W^2, \mu_i^p, \sigma_i^{p2}, \mu_S^p, \sigma_S^{p2}, a, b$) are treated as externally supplied constants, to be estimated from historical market data and meteorological records prior to simulation. In extended formulations, historical wind speed and solar irradiance profiles may be used to replace synthetic draws with empirical samples, preserving temporal autocorrelation and cross-variable dependence structures that the baseline model abstracts away.

Section 7 — Submodels

7.1 Solar Bid Quantity Profile

The solar producer's bid quantity for interval t is given by the truncated cosine function:

$$q_{S,t}^{\text{bid}} = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right)$$

The parameter $b > 0$ controls the amplitude of the diurnal cycle and a controls the baseline. When $a < b$, the profile produces zero-quantity bids in overnight intervals, as expected for a solar generator. The function is evaluated deterministically once per run for all $t \in \mathcal{T}$.

7.2 Merit-Order Stack Construction

Given all submitted bids $\{(q_{i,t}^{\text{bid}}, p_{i,t}^{\text{bid}})\}_{i \in \mathcal{I}}$ and the demand realisation D_t , the market operator constructs the merit-order stack π_t as follows:

1. Sort producers in non-decreasing order of $p_{i,t}^{\text{bid}}$.
2. Within any price tier where $p_{i,t}^{\text{bid}} = p_{j,t}^{\text{bid}}$ for $i \neq j$, apply Π_{priority} to determine relative rank.
3. The resulting sequence $\pi_t = (\pi_t(1), \dots, \pi_t(5))$ is the dispatch priority ordering for interval t .

7.3 Marginal Producer Identification and Price Setting

Define cumulative supply:

$$\mathcal{Q}_t(k) = \sum_{j=1}^k q_{\pi_t(j),t}^{\text{bid}}, \quad \mathcal{Q}_t(0) \equiv 0$$

The marginal producer rank and clearing price are:

$$m_t = \min\{k \in \{1, \dots, 5\} \mid \mathcal{Q}_t(k) \geq D_t\}, \quad \lambda_t = p_{\pi_t(m_t),t}^{\text{bid}}$$

If $\mathcal{Q}_t(5) < D_t$, the feasibility condition is violated; the supply-shortfall rule is invoked and $\delta_t = 0$ is recorded.

7.4 Dispatch Allocation

$$q_{\pi_t(j),t}^{\text{disp}} = \begin{cases} q_{\pi_t(j),t}^{\text{bid}} & j < m_t \\ D_t - \mathcal{Q}_t(m_t - 1) & j = m_t \\ 0 & j > m_t \end{cases}$$

By construction, $D_t - \mathcal{Q}_t(m_t - 1) \in (0, q_{\pi_t(m_t),t}^{\text{bid}}]$, so the marginal producer is dispatched at positive quantity and never over-dispatched.

7.5 Wind Delivery and Penalty Assessment

Following market clearing, actual wind production is drawn:

$$\tilde{A}_{W,t} \sim \mathcal{N}(\mu_W, \sigma_W^2)$$

The delivery shortfall and net revenue are:

$$s_{W,t} = \left(q_{W,t}^{\text{disp}} - \tilde{A}_{W,t} \right)^+$$

$$r_{W,t}^{\text{net}} = q_{W,t}^{\text{disp}} \cdot \lambda_t - \kappa \cdot s_{W,t}$$

Surplus production ($\tilde{A}_{W,t} > q_{W,t}^{\text{disp}}$) is curtailed without additional revenue. Total daily net revenue:

$$R_W^{\text{net}} = \sum_{t=1}^{24} r_{W,t}^{\text{net}}$$

7.6 Regulator Observation Submodel

The regulator computes the following derived quantities after all 24 intervals have cleared and been delivered:

$$\phi_t = \frac{q_{S,t}^{\text{disp}} + q_{W,t}^{\text{disp}}}{D_t}, \quad \bar{\lambda} = \frac{1}{24} \sum_{t=1}^{24} \lambda_t, \quad \bar{\phi} = \frac{1}{24} \sum_{t=1}^{24} \phi_t, \quad S_W^{\text{total}} = \sum_{t=1}^{24} s_{W,t}$$

These scalar outputs, together with the interval-level sequences $\{\lambda_t, \phi_t, \delta_t, s_{W,t}\}$, are the regulator's observation vector for a single simulation run and serve as the basis for evaluating system performance across N runs.

7.7 Wind Producer Optimisation Submodel

In wind-producer mode, the wind producer selects bid quantities and prices to solve:

$$\max_{\{q_{W,t}^{\text{bid}}, p_{W,t}^{\text{bid}}\}_{t \in \mathcal{T}}} \mathbb{E}[R_W^{\text{net}}] = \mathbb{E} \left[\sum_{t=1}^{24} \left(q_{W,t}^{\text{disp}} \cdot \lambda_t - \kappa \cdot \left(q_{W,t}^{\text{disp}} - \tilde{A}_{W,t} \right)^+ \right) \right]$$

subject to $q_{W,t}^{\text{bid}} \in [0, \bar{Q}_W]$ for all $t \in \mathcal{T}$, with κ and Π_{priority} treated as exogenous constants. The expectation is taken over all stochastic variables: D_t , competitor bid prices, and $\tilde{A}_{W,t}$. Since $q_{W,t}^{\text{disp}}$ depends non-linearly on $q_{W,t}^{\text{bid}}$ through the merit-order mechanism, and since the clearing price λ_t is jointly determined by all bids and demand, this expectation does not admit a closed-form solution in general and is evaluated numerically via Monte Carlo simulation across N runs.

7.8 Regulator Penalty Calibration Submodel

In regulator mode, the penalty rate κ is treated as a decision variable. For a given κ , the wind producer's best-response bidding strategy is obtained from Submodel 7.7. The regulator then evaluates system-level outcomes across N simulation runs under this best-response strategy. The optimal penalty rate κ^* is the value that satisfies the regulatory objectives in Section 4.4 — specifically minimising $\mathbb{E}[S_W^{\text{total}}]$ subject to $\mathbb{E}[\bar{\phi}] \geq \phi^{\min}$ — given the wind producer's endogenous response. This constitutes the lower-level constraint in the bilevel structure of the composed model. In practice, κ^* is identified via parameter sweep or derivative-free optimisation over the simulation.

In Section 7.1, parameter b should be less than 0 so that the solar output profile peaks at noon and drops to 0 at midnight. Please refine your specification accordingly.

7.1 Solar Bid Quantity Profile (Revised)

The solar producer's bid quantity for interval t is given by the truncated cosine function:

$$q_{S,t}^{\text{bid}} = \max \left(0, a + b \cos \left(\frac{2\pi t}{24} \right) \right)$$

where $a > 0$ is a baseline offset and $b < 0$ is an amplitude parameter. The sign of b is essential to the correct behaviour of the profile and warrants explicit justification.

The cosine function $\cos(2\pi t/24)$ attains its maximum value of $+1$ at $t = 0$ (midnight) and its minimum value of -1 at $t = 12$ (noon). Since $b < 0$, the term $b \cos(2\pi t/24)$ is therefore most negative at midnight and least negative — equivalently, most positive — at noon. The profile $a + b \cos(2\pi t/24)$ consequently peaks at $t = 12$ with value $a - |b|$, and reaches its minimum at $t = 0$ and $t = 24$ with value $a - |b| \cdot (-1) \cdot (-1) = a + b = a - |b|$...

More precisely:

Hour t	$\cos(2\pi t/24)$	$a + b \cos(2\pi t/24)$
$t = 0$ (midnight)	$+1$	$a + b = a - b $
$t = 6$ (dawn)	0	a
$t = 12$ (noon)	-1	$a - b = a + b $
$t = 18$ (dusk)	0	a
$t = 24$ (midnight)	$+1$	$a + b = a - b $

The profile is symmetric about noon, peaks at $a + |b|$, and reaches its minimum at $a - |b|$ at midnight. For the profile to produce zero output at midnight — as required for a solar generator with no overnight generation — the following condition must hold:

$a + b \leq 0 \iff a \leq |b|$

For the profile to produce strictly positive output at noon, it suffices that $a + |b| > 0$, which holds whenever $a > 0$ and $b < 0$. Combining these two conditions, the physically meaningful parameter region is:

$0 < a \leq |b|, \quad b < 0$

The $\max(0, \cdot)$ operator enforces non-negativity across all intervals, ensuring that rounding errors or parameter configurations where $a + b < 0$ do not produce inadmissible negative bid quantities. The constants table in Section 5 is correspondingly updated: the entry for b is qualified as $b < 0$, denoting a negative amplitude parameter whose magnitude $|b|$ controls the depth of the diurnal cycle.

Now, please provide a modular Python implementation of the unified model, with each agent implemented as a separate class and a single interface function defined at the end.

The implementation runs cleanly. Here is a summary of how the code maps to the ODD specification.

Structure

The file contains six logical sections that mirror the ODD protocol.

Agent classes (Sections 2-4)

Class	ODD entity	Key responsibility
ConventionalProducer	§2.2.1	Fixed-quantity bids; stochastic prices
SolarProducer	§2.2.2	Cosine profile quantities ($b < 0$ enforced); stochastic prices
WindProducer	§2.2.3	Control variables (q^{bid}, p^{bid}); delivery shortfall and penalty; inner MC loop for expected-revenue evaluation
MarketOperator	§2.2.4	Merit-order clearing with priority tie-breaking; partial dispatch at margin
SystemRegulator	§2.2.5	Post-run observation submodel; penalty-rate sweep for calibration

Supporting components

- BidSet, ClearingResult, DeliveryResult, SimulationResult — lightweight dataclasses carrying the state variables from §2.2 across phases.
- _single_run() — the internal execution kernel that steps through Phases 0-4 in the process schedule from §3, keeping the interface function clean.

Interface (run_simulation)

The single public entry point accepts all constants and uncertain parameters as arguments and dispatches to one of three evaluation modes.

mode	Active controls	Exogenous inputs
'wind_producer'	$q_{W,t}^{bid}, p_{W,t}^{bid}$	$\kappa, \Pi_{priority}$
'regulator'	κ (swept over kappa_sweep)	wind bids fixed
'composed'	both	none — bilevel structure

Wind market abm
PY

Download

Please modify the following two parts in your implementation: (1) Remove the constraint on the solar profile baseline offset a ; and (2) The quantity profile is computed with $t = \text{np.arange}(\text{INTERVALS})$, but the range of t should be 1, 2, ..., 24.

Both changes are localised to the `SolarProducer.__init__` method. The two edits are made below.

Two targeted changes were made to `SolarProducer.__init__`, and nothing else was touched.

Change 1 — Remove the constraint on a . The `ValueError` guard `if a <= 0` is deleted. The parameter `a` is now unconstrained, consistent with the model specification which imposes no sign restriction on the baseline offset; the `max(0, ·)` operator in the profile already handles any negative values that result from the combination of `a` and `b`.

Change 2 — Correct the interval index range. `np.arange(INTERVALS)` produced $t \in \{0, 1, \dots, 23\}$, which is misaligned with the specification's $\mathcal{T} = \{1, 2, \dots, 24\}$. The replacement `np.arange(1, INTERVALS + 1)` produces the correct range. This matters for the profile shape: with the original range, $t = 0$ evaluated to $\cos(0) = +1$, placing the minimum at the first element of the array rather than at $t = 12$ (noon). With the corrected range, $t = 12$ evaluates to $\cos(2\pi \cdot 12/24) = \cos(\pi) = -1$, so the term $b \cos(2\pi t/24)$ reaches its least-negative value — and the profile its maximum — precisely at noon, as specified.

Wind market abm
PY

Download

